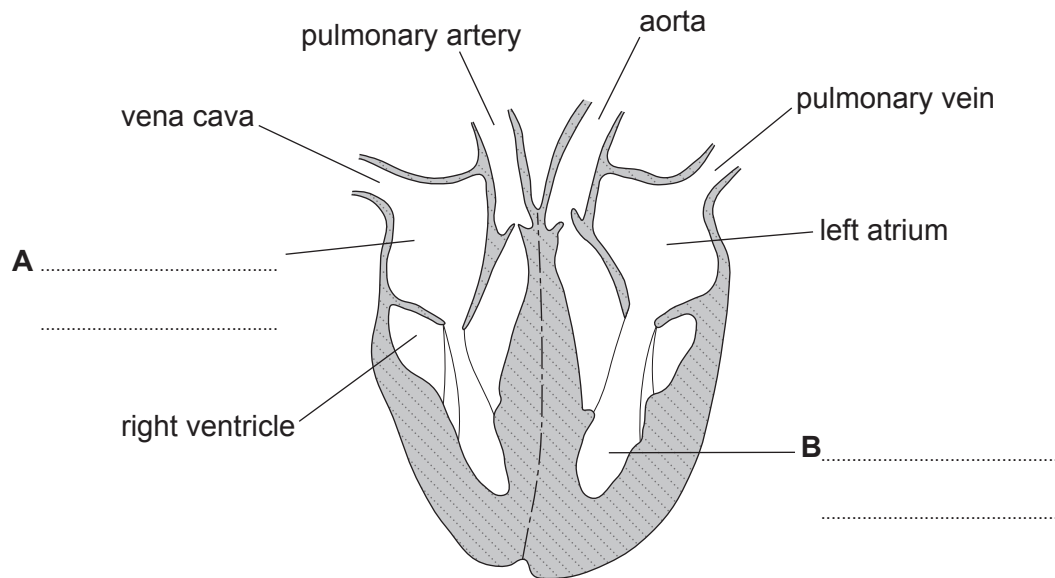


Answer **all** questions.

1. **Image 1** shows a section through the heart.

Image 1



- (a) Complete **Image 1** by adding the **two** missing labels (**A** and **B**) for the chambers of the heart. [2]
- (b) There are four main blood vessels connected to the heart.
Use a ruler to draw lines to match up each blood vessel with the correct function. [3]

Blood vessel	Function
pulmonary vein	transports blood from the body to the heart
aorta	transports blood from the heart to the body
vena cava	transports blood from the heart to the lungs
pulmonary artery	transports blood from the lungs to the heart



- (c) There are valves in the heart.
Underline the statement below that describes the function of the valves. [1]

Stop blood from clotting

Push blood through the heart

Prevent the backflow of blood

- (d) **Complete the following sentences**, using the correct words **from the list below**. [3]

pump muscle pulmonary carbon dioxide coronary

The heart is made of The function of the heart is to

..... blood around the body. The heart has its own blood supply

provided by the arteries.



Examiner
only

(ii) Describe **two** ways that the part of the respiratory system shown in **Image 2.1** is adapted for gas exchange. [2]

1.
2.

(c) **Table 2.2** shows the percentage of gases in inspired and expired air.

Table 2.2

Gas	Inspired air (%)	Expired air (%)
Nitrogen	78	
Oxygen	21	
Carbon dioxide		4

Complete Table 2.2 by adding the missing percentages to the empty boxes from the list below: [3]

0.04 16 78 21 4

10



- (iii) Use the results of both experiments to explain the advantage to the environment of adding enzymes to washing powder. [2]

.....

.....

.....

.....

- (d) Suggest **two** variables which should have been controlled in this method. [2]

1.
2.

